

Through the Iris: User Manual

A Gaze Estimation and Tracking System

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What This System Does

Through the Iris is a desktop application that tracks where you are looking on your screen using only your laptop or desktop webcam. No special hardware is needed.

The system watches your face through the camera, detects your eyes, and figures out which part of the screen you are looking at. It then displays this information in real time, drawing a small crosshair or trail on screen to show your current gaze position.

Beyond simple tracking, the application also collects data about your gaze patterns during a session and presents it as a heatmap, a zone map, and a blink rate graph, so you can see which parts of the screen held your attention most.

Who Is This For?

The system was built with a wide range of users in mind:

- Researchers studying attention or reading behaviour
- Students and educators interested in human-computer interaction
- Developers building gaze-aware applications
- Anyone curious about where their eyes go during everyday computer use

How It Works: The Big Picture

Understanding what happens under the hood can help you use the system better and interpret your results more accurately.

Step 1: The camera captures a video frame.

Your webcam continuously takes photos of your face many times per second.

Step 2: The system finds your face in the frame.

A face detection model (built on OpenCV's DNN tools) locates your face in the image and crops just that region.

Step 3: The face crop is standardised.

The cropped face image is resized to 224 by 224 pixels and normalised so that lighting and colour differences do not confuse the gaze model.

Step 4: A deep learning model predicts where you are looking.

This is the core of the system. A ResNet-50 neural network takes the face image and outputs two numbers: a pitch angle (up/down) and a yaw angle (left/right). Together, these define a direction in 3D space, which is then converted to a position on your screen.

Step 5: The prediction is smoothed.

Raw predictions jump around slightly from frame to frame. An Exponential Moving Average filter blends recent predictions together, producing a steadier gaze point.

Step 6: Calibration correction is applied (if calibration has been done).

If you have completed a calibration session, a personal correction is added on top of the model's prediction to account for your specific face shape, eye anatomy, and camera placement.

Step 7: The gaze point is displayed and recorded.

The corrected gaze position is shown on screen and added to the session data used by the dashboard.

Getting Started

First Launch

When you open the application for the first time, it will check for the required model files. If they are missing, it will download them automatically. This takes a few minutes depending on your connection speed.

Once the files are in place, the splash screen appears briefly while the model loads into memory.

The Sidebar

Every screen in the application has a vertical sidebar on the left side. This sidebar contains five buttons:

Icon	Screen
House	Home
Chart	Dashboard
Crosshair	Calibration
Gear	Settings
Info	About

Click any button to switch screens. Your session data and settings are not lost when you navigate between screens.

The Application Screens

Splash Screen

The splash screen appears while the application is loading. It displays the Through the Iris logo and name. You do not need to do anything here. The application will proceed automatically once it is ready.

Home Screen

The home screen is where you spend most of your time. It has two main areas.

Left side: Camera feed

This shows the live image from your webcam. When gaze tracking is active, a gaze trail appears on top of the camera feed, showing where your eyes have been over the last several frames.

Right side: Information panel

This panel shows:

- **Model status** (loaded or not loaded)
- **Camera status** (active or off)
- **Gaze X and Y coordinates** (where on screen you are looking, from 0 to 1)
- **FPS** (frames per second being processed)
- **Session duration** (how long the camera has been running)

Starting and stopping the camera

Click "Start Camera" at the bottom of the screen to begin tracking. Click "Stop Camera" to end the session. Your session data is preserved in the dashboard even after you stop.

Calibration Screen

Calibration personalises the gaze model to your specific face and camera setup. It takes about 20 seconds and is strongly recommended before use.

What the screen shows before calibration:

A simple instruction panel with three pieces of information: the number of calibration points (9), the time spent per point (roughly 1.5 seconds), and a "Begin Calibration" button.

What happens during calibration:

A full-screen black overlay appears with red dots arranged in a 3 by 3 grid. The dots appear one at a time. For each dot, you:

1. Look directly at the dot
2. Hold your gaze steady for about 1.5 seconds
3. Wait for the dot to blink, which means your sample was captured
4. Move your eyes to the next dot when it appears

You do not need to move your head. Keep your face roughly still and just move your eyes.

What happens after calibration:

The system uses the 9 samples to adjust the final layer of the neural network specifically for you. This process takes a few seconds. A results screen then shows your per-point accuracy and an overall score.

Recalibrating:

You can recalibrate at any time by clicking "Begin Calibration" again. Each calibration session overwrites the previous one.

Important: Calibration does not save between sessions. When you close and reopen the application, you will need to calibrate again. (Instructions for making calibration persist are in the advanced notes at the end of this manual.)

Dashboard Screen

The dashboard shows what your gaze has been doing over the course of a session. It does not require any interaction. Data flows in automatically as you use the home screen.

Session statistics row (top of screen):

- **Session duration:** Total time the camera has been active
- **Total blinks:** Number of blinks detected
- **Average FPS:** Frame processing rate for the session
- **Dominant zone:** The region of the screen you looked at most

Gaze heatmap:

A visual map of your screen where brighter, warmer colours indicate positions you looked at more. Areas you never looked at appear dark. The heatmap builds up gradually over the session. Up to 2,000 gaze points are stored at a time.

Gaze zone map:

Your screen is divided into a 3 by 3 grid of nine zones. Each cell shows the percentage of your gaze time spent in that region. Darker cells received more attention. This gives a quick, at-a-glance summary of whether you were focused in the centre, scrolling with your eyes to the right, etc.

Settings Screen

The settings screen lets you control how the system behaves. Changes are not saved until you click "Save Settings." An indicator at the top of the screen reminds you when there are unsaved changes.

Gaze Trail section:

- **Show Gaze Trail:** Toggle the gaze trail overlay on the home screen on or off.
- **Trail Length:** How many past gaze points appear in the trail. More points make a longer, slower-fading trail.

FPS Limit section:

- **Enable FPS Limit:** Caps the frame processing rate to save CPU.
- **FPS Cap:** The maximum frames per second to process when limiting is active.

Reset to Defaults: Returns all settings to their factory values.

About Screen

The About screen shows information about the project: the team, the institution, the technology stack, and a link to the repository. It also has a "Download Guide" button for the PDF version of this manual.

Frequently Asked Questions

Does the system record or upload my video?

No. All processing happens locally on your machine. No video, images, or gaze data are stored beyond the current session or sent anywhere.

Why is the gaze point sometimes in the wrong place?

Several factors can cause this: uncompleted calibration, poor lighting, an unusual camera angle, or the system detecting the wrong face in a cluttered background. Try recalibrating and ensuring good lighting.

What is FPS and how does it affect tracking?

FPS stands for frames per second, the rate at which the system processes camera images. Higher FPS means more frequent gaze updates and a smoother trail. If your CPU is struggling, enabling the FPS Limit in Settings will reduce CPU usage at the cost of slightly less frequent updates.